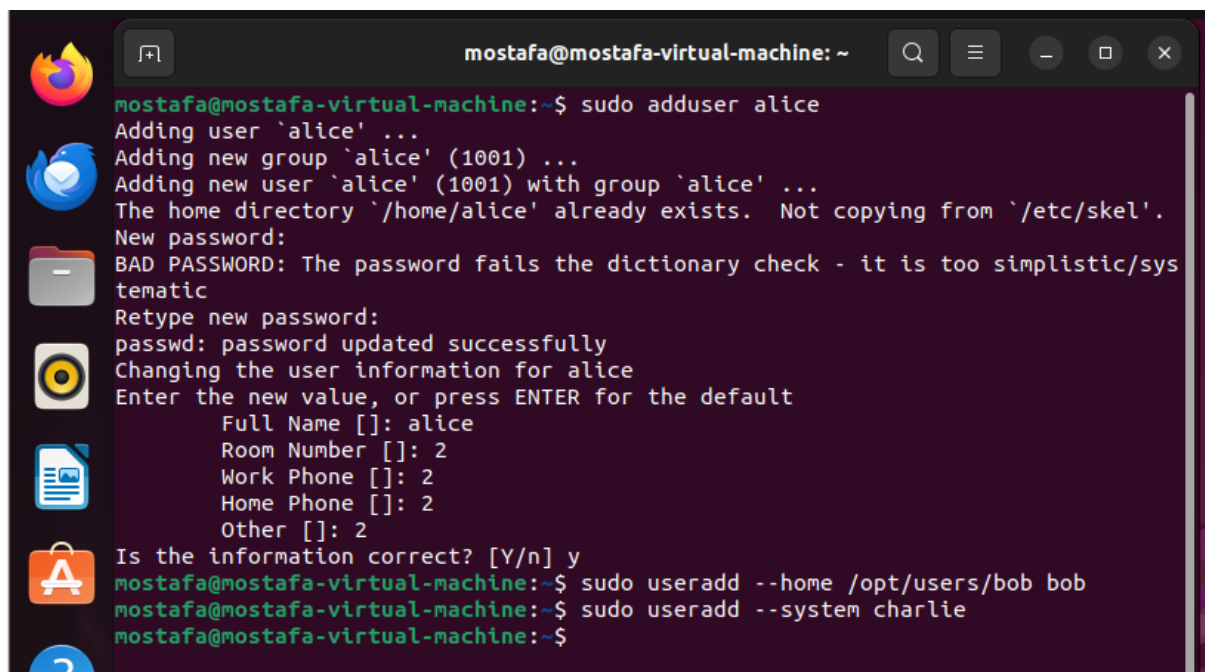


Name : Mostafa Ehab Diab Nassar

### ### Part A: User Creation and Management

1. **\*\*Create three new users\*\*** using ``adduser`` command:
  - alice
  - ``bob`` (with custom home directory ``/opt/users/bob``)
  - ``charlie`` (system user for services)



```
mostafa@mostafa-virtual-machine: ~  
mostafa@mostafa-virtual-machine:~$ sudo adduser alice  
Adding user `alice' ...  
Adding new group `alice' (1001) ...  
Adding new user `alice' (1001) with group `alice' ...  
The home directory `/home/alice' already exists. Not copying from `/etc/skel'.  
New password:  
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic  
Retype new password:  
passwd: password updated successfully  
Changing the user information for alice  
Enter the new value, or press ENTER for the default  
  Full Name []: alice  
  Room Number []: 2  
  Work Phone []: 2  
  Home Phone []: 2  
  Other []: 2  
Is the information correct? [Y/n] y  
mostafa@mostafa-virtual-machine:~$ sudo useradd --home /opt/users/bob bob  
mostafa@mostafa-virtual-machine:~$ sudo useradd --system charlie  
mostafa@mostafa-virtual-machine:~$
```

2. **\*\*Set passwords\*\*** for alice and bob:
  - Use ``passwd`` command
  - Force alice to change password at next login
3. **\*\*View user information\*\***:
  - Display alice's entry from ``/etc/passwd``
  - Show bob's password aging information using ``chage -l bob``

```

mostafa@mostafa-virtual-machine:~$ sudo passwd alice
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: password updated successfully
mostafa@mostafa-virtual-machine:~$ sudo passwd bob
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: password updated successfully
mostafa@mostafa-virtual-machine:~$ sudo passwd -e alice
passwd: password expiry information changed.
mostafa@mostafa-virtual-machine:~$ grep alice /etc/passwd
alice:x:1001:1001:alice,2,2,2,2:/home/alice:/bin/bash
mostafa@mostafa-virtual-machine:~$ sudo chage -l bob
Last password change                : 2025 ,19 أگ
Password expires                     : never
Password inactive                    : never
Account expires                      : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
mostafa@mostafa-virtual-machine:~$

```

### ▼ ### Part B: Group Management

#### ▼ 4. **\*\*Create two groups\*\*:**

- `developers` (regular group)
- `project-team` (using addgroup command)

#### ▼ 5. **\*\*Add users to groups\*\*:**

- Add alice to both `developers` and `project-team`
- Add bob to `developers` only
- Make `developers` alice's primary group

#### ▼ 6. **\*\*Verify group memberships\*\*:**

- Use `groups` command to show alice's groups
- Use `id` command to display bob's UID and GID information

```

mostafa@mostafa-virtual-machine:~$ sudo groupadd developers
mostafa@mostafa-virtual-machine:~$ sudo addgroup project-team
Adding group `project-team' (GID 1004) ...
Done.
mostafa@mostafa-virtual-machine:~$ sudo usermod -aG developers,project-team alice
mostafa@mostafa-virtual-machine:~$ sudo usermod -aG developers bob
mostafa@mostafa-virtual-machine:~$ sudo user -g developers alice
sudo: user: command not found
mostafa@mostafa-virtual-machine:~$ sudo usermod -g developers alice
mostafa@mostafa-virtual-machine:~$ groups alice
alice : developers project-team
mostafa@mostafa-virtual-machine:~$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),1003(developers)

```

```

2
3 ~ ### Part C: File Permissions and Ownership (25 points)
4
5 ~ 7. **Create test files and directories**:
6 |   - Create directory `/tmp/lab1_test`
7 |   - Create file `/tmp/lab1_test/data.txt` with some content
8 |   - Create file `/tmp/lab1_test/script.sh`
9
10 ~ 8. **Set permissions using symbolic notation**:
11 |   - Set `data.txt` permissions to `rw-r--r--` (owner: read/write, group: read, others: read)
12 |   - Set `script.sh` permissions to `rwxr-xr-x` (owner: full access, group/others: read/execute)
13
14 ~ 9. **Change ownership**:
15 |   - Change owner of `data.txt` to alice and group to developers
16 |   - Change owner of entire `/tmp/lab1_test` directory to bob and group to project-team (recursive)
17

```

```

mostafa@mostafa-virtual-machine:~$ sudo mkdir /tmp/lab1_test
mostafa@mostafa-virtual-machine:~$ echo "test " |sudo tee /tmp/lab1_test/data.txt
test
mostafa@mostafa-virtual-machine:~$ sudo touch /tmp/lab1_test/script.sh
mostafa@mostafa-virtual-machine:~$ sudo chmod 644 /tmp/lab1_test/data.txt
mostafa@mostafa-virtual-machine:~$ sudo chmod 755 /tmp/lab1_test/script.sh
mostafa@mostafa-virtual-machine:~$ sudo chown alice:developers /tmp/lab1_test/data.txt
mostafa@mostafa-virtual-machine:~$ sudo chown -R bob:project-team /tmp/lab1_test

```

### ### Part D: Verification and Testing

#### 10. \*\*Test permissions\*\*:

- Switch to alice user (`su - alice`)
- Try to read `data.txt`
- Try to execute `script.sh`
- Document what works and what doesn't

#### 11. \*\*Display final state\*\*:

- Use `ls -la /tmp/lab1\_test` to show all permissions and ownership
- Show group memberships for all created users

```
Activities Terminal 20:54 19 أگست
alice@mostafa-virtual-machine: ~
mostafa@mostafa-virtual-machine:~$ sudo su - alice
You are required to change your password immediately (administrator enforced).
Changing password for alice.
Current password:
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
New password:
Retype new password:
alice@mostafa-virtual-machine:~$ cat /tmp/lab1_test/data.txt
test
alice@mostafa-virtual-machine:~$ bash /tmp/lab1_test/script.sh
alice@mostafa-virtual-machine:~$ ls -la /tmp/lab1_test
total 12
drwxr-xr-x 2 bob project-team 4096 18:27 19 أگست .
drwxrwxrwt 20 root root 4096 18:24 19 أگست ..
-rw-r--r-- 1 bob project-team 6 18:26 19 أگست data.txt
-rwxr-xr-x 1 bob project-team 0 18:27 19 أگست script.sh
alice@mostafa-virtual-machine:~$ groups alice
alice : developers project-team
alice@mostafa-virtual-machine:~$ groups bob
bob : bob developers
alice@mostafa-virtual-machine:~$ groubs charlie
Command 'groubs' not found, did you mean:
  command 'groups' from deb coreutils (8.32-4.1ubuntu1.2)
Try: apt install <deb name>
alice@mostafa-virtual-machine:~$ groups charlie
charlie : charlie
alice@mostafa-virtual-machine:~$
```